

Subalgebra of the General Linear Algebra

1.1 Nilpotent Maps

Lemma 1.1.1

Let $x \in L$. If the linear map $x : V \rightarrow V$ is nilpotent, then $\text{ad } x : L \rightarrow L$ is also nilpotent.

Proof. Take any $y \in L$, and expand

$$(\text{ad } x)^m(y) = [x, [x, \dots [x, y] \dots]]$$

which is of the form

$$x^i y x^j, \quad i + j = m$$

Suppose that $x^r = 0$, taking $m = 2r$, we have $(\text{ad } x)^m(y) = 0$ for all $y \in L$. $\square$

Lemma 1.1.2

Suppose that A is an ideal of a Lie subalgebra L of $\mathfrak{gl}(V)$. Let

$$W = \{v \in V : a(v) = 0 \text{ for all } a \in A\}.$$

Then W is an L -invariant subspace of V .

Remark.

我们熟知, 如果每个 a 和 y 可交换, 则 W 一定是 L 不变的. 因此 W 成为 L 不变子空间, 只在于是否在 $[a, y]$ 下保持, 这也是为什么我们要假设 A 是一个理想.

Proof. Take $w \in W$ and $y \in L$, we must show that $a(yw) = 0$ for all a . But $ay = ya + [a, y]$, we have

$$a(yw) = y(aw) + [a, y](w)$$

And $y(aw) = 0$ for $aw = 0$; $[a, y](w) = 0$ for $[a, y] \in A$. □

1.2 Weights

Definition 1.2.1

A **weight** for a Lie subalgebra A of $\mathfrak{gl}(V)$ is a linear map $\lambda : A \rightarrow F$ such that

$$V_\lambda := \{v \in V : a(v) = \lambda(a)v \text{ for all } a \in A\}$$

is a non-zero subspace of V .

1.3 Invariance Lemma

Lemma 1.3.1

Suppose that A is an ideal of a Lie subalgebra L of $\mathfrak{gl}(V)$. Let

$$W = \{v \in V : a(v) = 0 \text{ for all } a \in A\}.$$

Then W is an L -invariant subspace of V .

Lemma 1.3.2 ► Invariance Lemma

Assume that F has characteristic zero. Let L be a Lie subalgebra of $\mathfrak{gl}(V)$ and let A be an ideal of L . Let $\lambda : A \rightarrow F$ be a weight of A . The associated weight space

$$V_\lambda = \{v \in V : av = \lambda(a)v \text{ for all } a \in A\}$$

is an L -invariant subspace of V .